Parametric APDL Horizontal Tailplane Structural Analysis

Project Schedule & Milestone Tracking

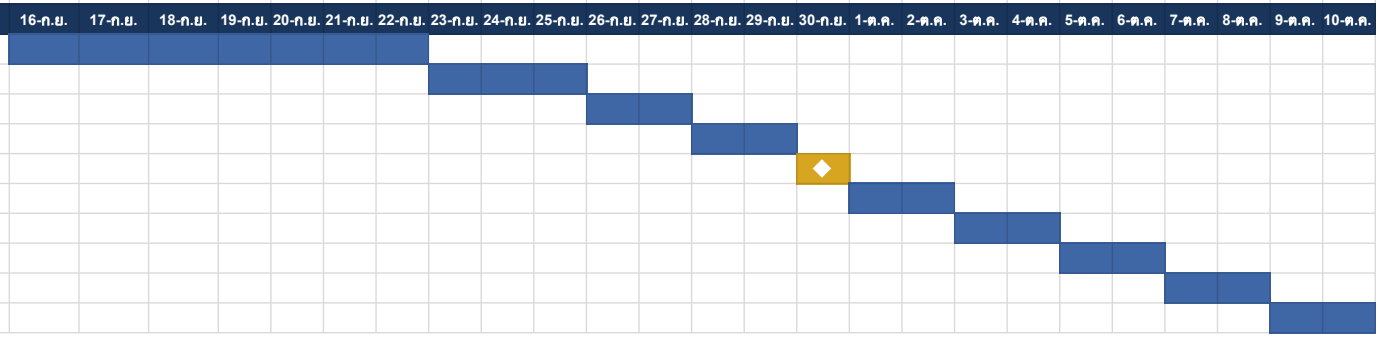
Total Tasks	In Progress	Pending	Proposal Milestone
10	1	6	2026-09-30

Task ID	Task Name	Category / Deliverable	Start Date	End Date	Duration (Days)
T-01	Initial topic study (aircraft window structure)	Topic Selection & Literature	2026-09-16	2026-09-22	7
T-02	Topic change to horizontal tailplane; literature review	Topic Selection & Literature	2026-09-23	2026-09-25	3
T-03	Step 1: Data collection & parameter definition	Stage 1: Input Data	2026-09-26	2026-09-27	2
T-04	Step 2: Parametric APDL script development	Stage 2a: APDL Script	2026-09-28	2026-09-29	2
T-05	Proposal presentation for project approval	Major Milestone	2026-09-30	2026-09-30	1
T-06	Step 3: SFBM load pre-processing (MATLAB)	Stage 2b: Load Reconstruction	2026-10-01	2026-10-02	2
T-07	Step 4: Configuration setup & Step 5: Model verification	Stage 3a-b: Setup & Verification	2026-10-03	2026-10-04	2
T-08	Step 6: Static analysis & Step 7: Baseline validation	Stage 3c: Static FEM Analysis	2026-10-05	2026-10-06	2
T-09	Step 8: Evaluation & comparison (Config A, B, C)	Stage 3d: Optimization	2026-10-07	2026-10-08	2
T-10	Step 9: Final Report preparation	Documentation & Submission	2026-10-09	2026-10-10	2
Total					**25**

Horizontal Tailplane Project Timeline (Gantt Chart)

Visual Project Workflow | Aerospace Structural Optimization

Task ID	Task Name
T-01	Initial topic study (aircraft window structure)
T-02	Topic change to horizontal tailplane; literature review
T-03	Step 1: Data collection & parameter definition
T-04	Step 2: Parametric APDL script development
T-05	Proposal presentation for project approval
T-06	Step 3: SFBM load pre-processing (MATLAB)
T-07	Step 4: Configuration setup & Step 5: Model verification
T-08	Step 6: Static analysis & Step 7: Baseline validation
T-09	Step 8: Evaluation & comparison (Config A, B, C)
T-10	Step 9: Final Report preparation



Estimated Project Budget & Resource Allocation

Coursework & Laboratory Resource Utilization | APDL & MATLAB

Total Allocated Budget	Total Actual Cost	Net Variance
THB 0.00	THB 0.00	THB 0.00

Item / Resource	Description	Source & Licensing	Cost (THB)	Funding Status
ANSYS Mechanical APDL 2026 R1	Finite element modelling and structural analysis	ANSYS Student version (free)	THB 0.00	University / Academic Free
MATLAB	SFBM load pre-processing & numerical solver	University license (student academic account)	THB 0.00	Institutional License
Computer workstations	Running simulations and mesh generation	Personal student laptops	THB 0.00	Existing Personal Resource
Course materials & reports	Documentation and presentation materials	Digital submission	THB 0.00	Department Digital Facility
Total Estimated Cost			**THB 0.00**	**Fully Funded / Zero Cost**